

# FINEXA

## Smart Loan Management System

Complete System, Product & Technical Documentation

### DOCUMENT CONTROL & SPECIFICATIONS

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Repository: <https://github.com/ponnagantiharikrantsai-tech/Finexa>

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20. [REDACTED] Design

## 1. EXECUTIVE SUMMARY

FINEXA is a production-grade FinTech Loan Management System engineered for private lenders, financial administrators, and credit management teams. FINEXA replaces error-prone spreadsheets with a secure, real-time platform for issuing loans, calculating monthly interest yields, enforcing payment cycles, tracking late penalties, and generating compliance ledgers.

- Primary Purpose: Lifecycle management of loan principal, monthly interest, cycle extensions, and overdue penalty recovery.
- Target Audience: Private lenders, loan officers, administrative teams, and portfolio auditors.
- Core Innovation: Instant 60 FPS response, multi-device real-time state sync, and strict server-side security boundaries.

## 2. PROJECT OBJECTIVE

The principal objective of FINEXA is to provide a single, immutable source of truth for all loan issuing, borrower KYC records, repayment ledgers, and financial calculations.

- Borrower KYC: AES-256-CBC encrypted storage of Aadhaar, PAN, identity details, and location URLs.
- Loan Cycle Precision: Standardized calculation of monthly interest due dates, extensions, and late fee penalties.
- Pay & Extend Standardization: Standardized workflow when borrowers pay monthly interest without principal reduction.
- Overdue Penalty Settlement: Dynamic daily penalty accrual per  $\frac{1}{1,000}$  principal / day.

### 3. COMPLETE APPLICATION FLOW

The application flow establishes a seamless path from entry to compliance archiving:

## MASTER SYSTEM NAVIGATION ARCHITECTURE

Public Visitor -> Public Landing & Login (/login) -> Authenticated Session

```
%% > Home Dashboard (/home)
```

```
%%%> Loan Management (/loan-management) -> Issue Loan -> Active Card -> Pay & Ext
```

%%% > Borrower Management (/borrowers) -> View Details -> Encrypted KYC -> Internal

```

%%% > Application Generator (/applications) -> Public Apply Link -> Submission -> Verif

```

%%> Reports & Analytics (/reports) -> Financial Ledgers -> CSV Export / Print PDF

Capital Management (/capital-management) -> Vault Balance -> Funder Deposits

## 4. USER ACCESS FLOW & PUBLIC/PRIVATE BOUNDARY

FINEXA enforces a strict perimeter between public landing pages and private financial records:

- PUBLIC ROUTE: /login & / -> Accessible without credentials. Displays lightweight login interface. Indexable by search engines.
- PUBLIC ROUTE: /apply/[code] -> Customer borrower application link.
- PRIVATE ROUTES: /home, /loan-management, /borrowers, /reports, /capital-management, /settings -> Protected by Next.js middleware.
- Middleware Enforcement: Server-side redirection to /login if unauthenticated. Protected routes stamp Cache-Control: no-store headers to prevent browser back-button data leakage upon logout.

## 5. HOME PAGE ARCHITECTURE

The Home Dashboard (/home) provides executive overview access to all core operational modules:

- 3D Hero Branding: Perspective-aware FINEXA logo depth effect.
- Staggered Navigation Cards: Direct access to Loan Management, Applications, and Financial Reports.
- Instant Data Hydration: Renders interface shell immediately without blocking on secondary data queries.

## 6. LOAN MANAGEMENT WORKFLOW

Loan Management (/loan-management) is the central operational module of FINEXA:

- New Loan Creation: Select existing borrower or inline create new borrower card with encrypted KYC.
- Loan Terms: Principal (<sup>1</sup>), Monthly Interest Rate (<sup>1</sup> per <sup>1</sup>1,000 / month), Date Given, Due Date.
- Active Card Summary: Displays Principal, Outstanding Balance, Monthly Interest, Current Due Date, and Status (active, extended, overdue, closed).

## 7. BORROWER MANAGEMENT SYSTEM

Borrowers (/borrowers) manages complete identity and credit history:

- Encrypted KYC: AES-256-CBC encrypted storage for PAN and Aadhaar records.
- Card Association: Links every borrower card to their respective active and historical loans.
- Collision Resolver: Unique mobile resolver prevents database constraint failures when creating separate borrower cards for identical mobile numbers.

## 8. INTEREST SYSTEM LOGIC

FINEXA calculates monthly interest yields using standard FinTech lending formulas:

### FINTECH INTEREST FORMULA SPECIFICATION

Monthly Interest Formula:

Monthly Interest = Principal \* (Interest Rate / 1,000)

Example:

Principal = <sup>1</sup>10,000 | Interest Rate = <sup>1</sup>250 per <sup>1</sup>1,000 / month

Monthly Interest = 10,000 \* (250 / 1,000) = <sup>1</sup>2,500 / month

## 9. PAY & EXTEND WORKFLOW

Pay & Extend handles the common scenario where a borrower pays monthly interest without principal reduction:

- Interest Recorded: Interest payment (e.g. <sup>1</sup>2,500) is recorded into payments ledger.
- Principal Unchanged: Outstanding principal remains 100% (e.g. <sup>1</sup>10,000).

- Cycle Advanced: Current cycle closes as 'extended' and due date extends by 1 month.
- Audit Record: Permanent cycle entry inserted into loan\_cycles database table.

## 10. OVERDUE & PENALTY CALCULATION SYSTEM

When a borrower pays after the assigned due date, late penalty accrues dynamically:

### PENALTY ACCRUAL FORMULA SPECIFICATION

Accrued Penalty Formula:

$\text{Penalty} = (\text{Principal} / 1,000) * \text{Penalty Rate} * \text{Overdue Days}$

Example:

Principal = ₦10,000 | Rate = ₦2.50 per ₦1,000 / day | Overdue = 2 Days

$\text{Penalty} = (10,000 / 1,000) * 2.50 * 2 = ₦50$

Total Settlement Required = Interest (₦2,500) + Penalty (₦50) = ₦2,550

## 11. PAYMENT SYSTEM & ALLOCATION

Payments are recorded with explicit type allocation:

- Interest Payment: Satisfies monthly cycle interest requirement.
- Principal Payment: Directly reduces outstanding principal balance.
- Penalty Payment: Settles late fee charges recorded in penalty\_ledger.

## 12. PERMANENT LOAN CYCLE LEDGER SYSTEM

The loan\_cycles table stores an immutable audit history of every cycle transition:

- Attributes: cycle\_number, start\_date, due\_date, payment\_date, interest\_paid, penalty\_paid, remaining\_principal, status.
- Transparency: Full historical cycle timeline viewable inside Loan Details popup.

## 13. APPLICATION LINK GENERATOR WORKFLOW

Allows lenders to generate digital KYC application links for borrowers:

- Link Generation: Unique application code generated with pre-set principal and interest terms.
- Borrower Submission: Borrower opens /apply/[code], fills KYC details, submits.
- Verification: Lender reviews pending application in /applications and approves to auto-create loan & borrower.

## 14. REPORTS & FINANCIAL ANALYTICS

Reports (/reports) provides real-time financial health analytics:

- Metrics: Total Amount Lent, Total Collected, Total Outstanding, Interest Earned, Net Profit.
- Filtering: Today, Yesterday, Last 7 Days, Last 30 Days, This Month, Last Month, Custom Range.
- Exports: One-click Excel/CSV data export and print-ready compliance layout.

## 15. PDF GENERATION ARCHITECTURE

FINEXA features standalone PDF generation powered by jsPDF:

- Active Loans Backup PDF: Generates offline printable backup of every active loan card.
- Non-Blocking UI: Progress overlay displays while PDF compiles without freezing application.

## 16. FUNDER / CAPITAL MANAGEMENT

Capital Management (/capital-management) tracks investor vault balances:

- Capital Addition: Records additions to available loan capital pool.
- Loan Utilization: Tracks available vs deployed capital across active loans.

## 17. INTERNAL NOTES SYSTEM

Lenders can store private administrative notes per borrower:

- Auto-Save: Notes saved to database upon closing details view or clicking save.
- Audit Track: Note updates record internal\_notes\_updated\_at timestamp.

## 18. SEARCH, FILTERING & DEBOUNCING

Search and filtering are optimized for 60 FPS performance:

- Multi-Field Search: Matches Name, Mobile, PAN, and Aadhaar.
- Debounced Execution: Uses useDeferredValue and useMemo to prevent UI frame drops during fast typing.

## 19. MULTI-DEVICE REAL-TIME SYNCHRONIZATION

FINEXA utilizes Supabase Realtime subscriptions to reflect changes across devices instantly:

- Event Listening: Subscribes to INSERT, UPDATE, DELETE on loans and payments.
- Silent Hydration: Updates local UI state automatically without full page reload.

## 20. AUTHENTICATION & SECURITY ARCHITECTURE

Security is implemented across every application layer:

- Auth Provider: Supabase SSR Authentication with encrypted JWT cookies.
- Data Encryption: Sensitive PAN and Aadhaar encrypted at rest via AES-256-CBC.
- Anti-Caching: Protected pages set Cache-Control: no-store headers.

## 21. NOTIFICATION ARCHITECTURE (EMAIL/SMS)

Automates borrower communications:

- Email Notifications: Powered by Resend API & React Email templates.
- SMS Integration: Fast2SMS API integration for loan creation and overdue reminders.
- Status: Email (Implemented), SMS (Configured), WhatsApp (Not currently implemented).

## 22. DATABASE SCHEMA & ER DIAGRAM

PostgreSQL relational schema managed via Drizzle ORM:

### DATABASE ENTITY RELATIONSHIPS

```
Borrowers (borrower_id) %> 1:N %> Loans (loan_id)
%> 1:N %> Payments (payment_id)
%> 1:N %> Loan Cycles (cycle_id)
%> 1:N %> Penalty Ledger (penalty_id)
```

## 23. DATA FLOW ARCHITECTURE

Client Action -> Server Action -> Drizzle ORM -> Supabase PostgreSQL -> Realtime Dispatch -> Optimistic UI Update.

## 24. VERIFIED TECHNOLOGY STACK

- Framework: Next.js 16.2 (Turbopack, App Router, Server Actions)
- Language & UI: TypeScript 5, React 19, Tailwind CSS v4, Base UI
- Database & ORM: Supabase PostgreSQL, Drizzle ORM 0.45
- Security: AES-256-CBC Encryption, Supabase Auth SSR

## 25. DIRECTORY CODE STRUCTURE

Modular feature-based architecture under /src:

- src/app: Next.js App Router routes ((auth), (dashboard), api, apply).
- src/features: Domain modules (loans, borrowers, payments, applications, reports, capital).
- src/db: Schema definitions & Drizzle database client.
- src/components/motion: 3D Motion System components.

## 26. PAGE FEATURE MATRIX

- /login: Public login entrance animation and credentials submission.
- /home: Executive hero dashboard and module access.
- /loan-management: Loan card list, payments, extensions, penalties, PDF download.
- /reports: Financial analytics, KPIs, CSV export, print view.

## 27. BUTTON ACTION MATRIX

- Record Payment: Submits payment action -> Money particle effect -> Local state update.
- Pay & Extend: Executes interest clearance & cycle extension -> 3D timeline rotation.
- Download PDF: Compiles active loan records into downloadable backup PDF.

## 28. SYSTEM ERROR HANDLING STRATEGY

Form actions return typed `ActionResult<T>`. Errors display via toast notifications with clear diagnostics and state rollback.

## 29. 60 FPS 3D MOTION SYSTEM

Custom GPU-accelerated motion system utilizing CSS 3D transforms (`translateZ`) and HTML5 Canvas particles without heavy 3D engine overhead.

## 30. RESPONSIVE MOBILE/DESKTOP DESIGN

Fluid layouts designed for Mobile, Tablet, and Desktop with touch-friendly press interactions replacing hover on mobile viewports.

## 31. SEO & GOOGLE DISCOVERABILITY

Public entrance configured with metadata, OpenGraph, sitemap.xml, and robots.txt disallowing private route crawling.

### 32. BACKUP & RECOVERY PROTOCOL

Database auto-backups managed by Supabase Cloud. Source code version-controlled on GitHub.

### 33. CI/CD DEPLOYMENT PIPELINE

Git Push to main branch triggers automatic Vercel production build and deployment.

### 34. REAL-WORLD SCENARIO EXAMPLE

Step-by-step trace of loan creation, interest payment, extension, overdue penalty, and final closure.

### 35. MASTER SYSTEM FLOWCHART

Complete decision tree from user entry to loan payoff and compliance archiving.

### 36. GLOSSARY OF FINTECH TERMS

- Principal: Original loan amount issued.
- Interest Rate: Monthly yield per ₦1,000 principal.
- Pay & Extend: Interest clearance without principal reduction.

### 37. CURRENT VS FUTURE ROADMAP

- Current: Loan management, Pay & Extend, Penalties, PDF, Reports, 3D Motion, Realtime.
- Future: WhatsApp Business API reminders, automated CIBIL score checks (Planned).

### 38. FINAL LIFECYCLE SUMMARY

FINEXA delivers a complete, secure, fast, and visually striking loan management platform built to scale with growing lending operations.